

BESS RFP

UAE DEWA VII BESS Project

	Project Name	Doc. No.:			
Sponsor: Masdar	UAE DEWA VII BESS Project	Rev.	[●]	Std.	S-0.1

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1. ABBREVIATIONS

BESS	Battery Energy Storage System
BOO	Build Own Operate
COD	Commercial Operation Date
DEC	Default Equivalent Cycles
DoD	Depth of Discharge
EGP	Early Generation Period
ISP	Independent Storage Project
LFP	Lithium Ferro Phosphate
LTSA	Long Term Service Agreement
PoC	Point of Connection at HV
RTE	Round Trip Efficiency
SCOD	Scheduled Commercial Operation Date
SOH	State of Health

2. INTRODUCTION

Abu Dhabi Future Energy Company PJSC (“**Masdar**” or the “**Buyer**”) has been invited to submit a bid by Dubai Electricity & Water Authority (“**DEWA**” or the “**Offtaker**”) for the development, permitting, financing, design, engineering, procurement, construction, testing, commissioning, completion, insurance, ownership, operation and maintenance of a Hybrid Plant with 2GWh Solar PV and 1.4GWh BESS over six hours in DEWA VII (the “**Project**”).

The Project will be structured as an Independent Power Project and will be developed on a Building Own Operate “BOO” basis. The Project will be implemented in Phases with the scheduled COD as follows:

1. Phase A: 400 MWac PV (with 260 MW 6-hour BESS); Phase A SCOD by October 2027,
2. Phase B: 800 MWac PV (with 570 MW 6-hour BESS); Phase B SCOD by October 2028, and
3. Phase C: 800 MWac PV (with 570 MW 6-hour BESS); Plant COD by November 2029.

The Buyer has prequalified a limited number of BESS integrators (each a “**Bidder**” and collectively “**Bidders**”) based on previous experience and will be seeking the most competitive offer based on cost and commercial terms while complying with the requirements of this RFP and its Appendices.

3. SITE INFORMATION

The plant will be sited on a greenfield plot within the Mohammed Bin Rashid Al Maktoum Solar Park, in the Saih Al Dahal area approximately 50 km south of Dubai, as shown in the locality map in Figure 1. The Project will have a total area of approximately 20.34km². The site has an average elevation of 161 meters above sea level.

Severe climatic conditions apply at the site, including high ambient humidity at high air temperatures and extreme wind events such as sudden gusts and storms. Protection against high ambient temperatures, high ambient humidity, strong winds, dust, and sand for outdoor and indoor equipment shall receive special attention. Rainfall is highly erratic, and attention shall be paid to the possibility that sudden heavy downpours, severe storms, and flash floods may occur. Strong winds are common, and sandstorms and dust storms are likely.

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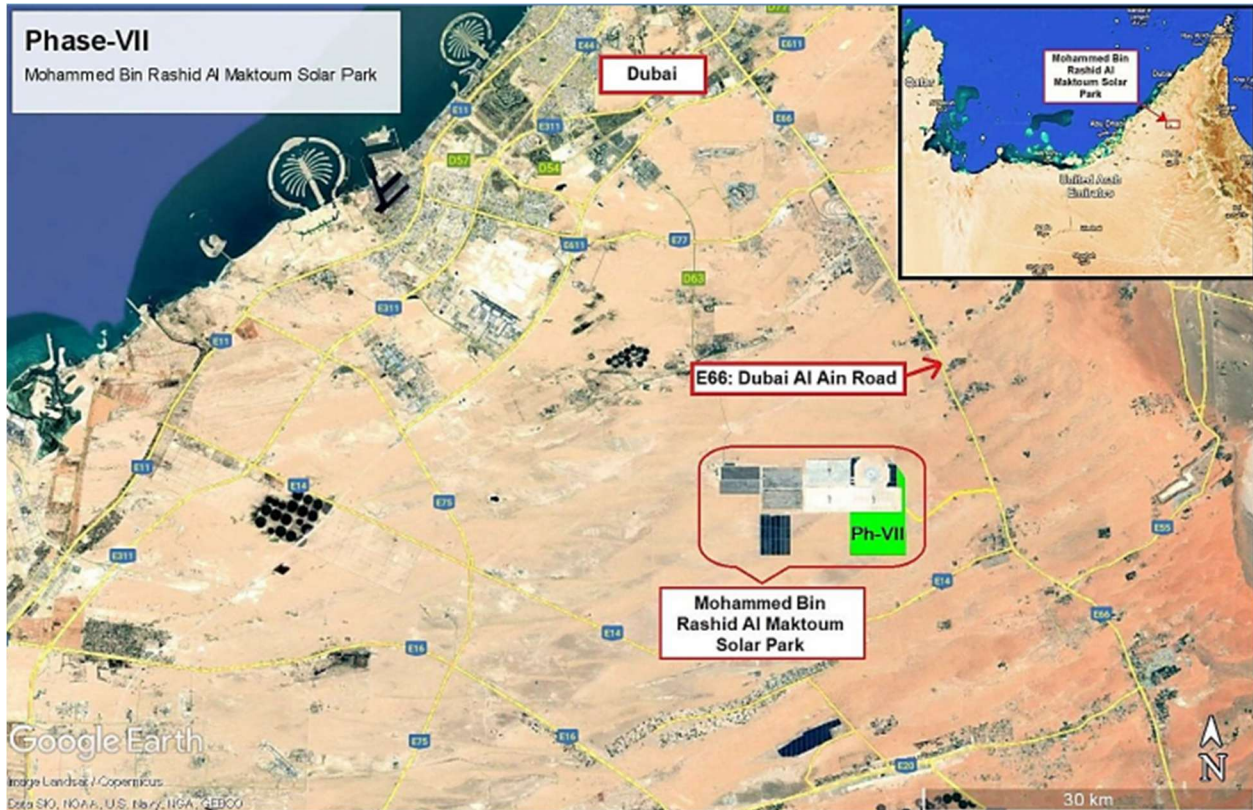


Figure 1: Location of the Mohammed Bin Rashid Al Maktoum Solar Park

The minimum design requirements for the Project shall be determined considering climatic conditions briefly described below.

Table 1: Minimum Design Climatic Conditions

Temperatures	Units	Value
Maximum ambient temperature in the shade, to be taken as air inlet temperature for air-conditioning and ventilation	°C	55
Minimum ambient temperature	°C	2.8
Maximum metal temperature in the sun	°C	85
Ambient temperature for outdoor installations (in the shade)	°C	55
Design temperature of electrical equipment (indoor)	°C	55

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Maximum design temperature for electrical/electronic equipment in air-conditioned rooms with N+1 redundancy (excluding BESS containers)	°C	25
Relative Humidity	Units	Value
Maximum ambient relative humidity	%	100
Average monthly ambient relative humidity	%	38-65
Average monthly minimum ambient relative humidity	%	11-33
Annual Rainfall	Units	Value
Average annual rainfall	mm	50
Maximum recorded rainfall in one day	mm	190
Wind	Units	Value
Maximum mean 10-min wind velocity at 10m height	m/s	35
Maximum 3 sec wind gust at 10 m height	m/s	53.5
Dust Concentration	Units	Value
Dust concentration in the air under normal conditions	mg/m3	1.0–20
Dust concentration in the air under sandstorm conditions	mg/m3	Approx. 100

4. SCOPE OF SUPPLY

Bidders are requested to propose the most advanced BESS solution available in their product roadmap while complying with the requirements of this RFP and its Appendices. The scope of supply shall consist of:

- Basic and detailed engineering.
- Procurement of material and equipment.
- Manufacturing and testing.
- Quality control.
- Protection and packaging for transport, shipment and storage at site.
- Supervision during installation and commissioning.

Bidder shall clarify components, OEMs, and product warranties of its solution in Appendix 2 BF #3 (Scope of Supply).

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5. ELIGIBILITY AND REQUIREMENTS

5.1. Eligibility

The BESS solution must be technically suitable for the site-specific conditions detailed in section 3. The BESS supplier is listed on the most recent Bloomberg NEF Tier 1 list of battery storage manufacturers.

The Bidder must also demonstrate the successful track records as follows:

- At least three (3) operational utility-scale BESS references, with an installed capacity of at least 50MWh, located in hot desert climates.
- Global delivery of at least 1,400MWh (DC) of storage capacity.

The reference BESS projects must be of a utility-scale nature, either as a standalone system or integrated with renewable or conventional power generation facilities. It must include functionalities related to grid support, such as frequency regulation and voltage control, and be connected either to a high-voltage (HV) transmission network or a microgrid.

In addition, the proposed model of battery modules and battery enclosure series shall have:

- deployment in at least two (2) BESS plants with an installed capacity of at least 100MWh each.
- a cumulative installed capacity of at least 1,000MWh.

The two reference sites ($\geq 100\text{MWh}$) and the accumulative installed capacity shall have been installed and successfully operated for at least one (1) year prior to Buyer's Proposal Submission Date.

The PCS manufacturer shall comply with the following requirements:

- At least five (5) years of PCS production.
- A minimum annual production capacity of 500MW.
- Have supplied PCS to at least one (1) BESS project with the DC installed capacity of at least 300MWh in the past 2 years, which have all been financed non-recourse by three different banks.

The Bidder must be fully incorporated and in valid existence at the time of proposal submission deadline.

5.2. Certifications

Bidder shall fulfill Appendix 2 BF #4 (Certifications) indicating which certifications are already available.

Bidder shall provide those third-party certificates as part of its proposal.

5.3. Track Record

Bidder shall fulfill Appendix 2 BF #5 (Track record) with its operational references in home country and overseas.

6. BASIS OF DESIGN

As specified in Appendix 2 BF #1 (Technical Parameters), particularly:

- Project lifetime: 25 years from Plant COD
- BESS lifetime: minimum 25 Years (please provide a solution with extended BESS lifetime up to 30 years for information purposes)
- BESS Plant DC installed energy capacity: to be proposed by Bidder as per the specific BESS solution

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- BESS Plant DC functional energy capacity @ PCOD: Bidder to calculate as per section 6.2.
- BESS Plant usable power capacity @ PoC (accounting for system losses):

Duration	BESS Plant usable power capacity @ PoC (accounting for losses)
Phase A SCOD – Phase B SCOD	260 MW for Phase A
Phase B SCOD – Plant COD	830 MW for Phase A and B
Plant COD till Project Lifetime	1,400 MW for Phase A, B, and C

- BESS Plant usable energy capacity @ PoC (accounting for system losses):

Duration	BESS Plant usable energy capacity @ PoC (accounting for losses)
Phase A SCOD – Phase B SCOD	1,560 MWh for Phase A
Phase B SCOD – Plant COD	4,980 MWh for Phase A and B
Plant COD till Project Lifetime	8,400 MWh for Phase A, B, and C

- Default equivalent cycles: 1 per day
- Rated P-Rate: 0.167 (or 1/6)
- Cell chemistry: LFP

The Bidder shall base Phase A and Phase B on the Bidder's latest commercially available BESS products at the Proposal Submission Date. For Phase C, the Bidder is free to offer BESS products scheduled for commercial release during 2027–2028, subject to: (i) compliance with all technical specifications and performance guarantees; (ii) compliance with the required certifications; (iii) no negative impact on Plant COD.

6.1. Technical Requirements

Bidders shall comply with the technical specification included in Appendix 1 and all of its attachments.

Any deviation to those specifications shall be proposed to the Buyer using Appendix 2 BF #8 (Declared Deviations).

Deviations outside Appendix 2 BF #8 (Declared Deviations), whether declared or not, will not be considered.

Additionally, Bidder shall provide Appendix 2 BF #1 (Technical Parameters) fulfilled in sufficient detail.

6.2. DC Installed Energy Capacity vs DC Functional Capacity

Parameters to be considered to derive the DC Functional Energy Capacity from the DC Installed Energy Capacity:

- DoD.
- Energy capacity loss due to voltage mismatch affecting the battery packs and racks energy capacity complete utilization.
- OEM design margin.

Assuming the difference between DC Functional Energy Capacity and DC Installed Energy Capacity is $\Delta\%$ (refer to Item #6.04 in BF #1 (Technical Parameters)), the DC Functional Energy Capacity shall be calculated as follows:

$$\text{DC Functional Energy Capacity (i)} = \text{DC Installed Energy Capacity} \times (1 - \Delta\%) \times \text{SOH (i)}$$

Where

- i indicates the specific Contract Year.
- SOH (i) is the SOH for Contract Year i.

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DoD shall not be considered as an efficiency loss occurring during the charge and discharge processes.

6.3. Charge, Discharge and RTE Efficiencies

The effective charge efficiency shall be calculated as the product of the efficiency factors listed in this section. The effective discharge efficiency shall be calculated similarly but shall exclude the cell efficiency factor. The cell efficiency factor is the only factor that varies over time due to cell aging and degradation. RTE shall be calculated by directly multiplying the charge and discharge efficiencies.

Even though the [HV cabling (PoC - MV/HV transformer)], [MV/HV transformer], MV cabling ([PoC or MV/HV transformer] - LV/MV transformer), LV cabling (LV/MV transformer - PCS) and the DC cabling (PCS - DC panel) are excluded from the Bidders scope of supply, as specified in Appendix 2 BF #3 (Scope of Supply), Bidders shall retain the responsibility of their design and sizing as per applicable local and international codes and standards ensuring alignment with the efficiency factors declared in Appendix 2 BF #1 (Technical Parameters).

Efficiency factors:

- [HV cabling (PoC - MV/HV transformer)]
- [MV/HV transformer]
- MV cabling ([PoC or MV/HV transformer] - LV/MV transformer)
- LV/MV transformer
- LV cabling (LV/MV transformer - PCS)
- PCS
- DC cabling (PCS - DC panel)
- DC cabling (DC panel - battery pack)
- DC/DC converter (if applicable)
- Cell.

Bidder shall declare the SOH, the cell efficiency, the charge efficiency and the effective discharge efficiency of its proposed solution in Appendix 2 BF #2 (Performance Metrics). The RTE is automatically calculated by Appendix 2 BF #2 (Performance Metrics) following the principles above. Formulas shall not be modified.

6.4. Energy Capacity Augmentation

Bidders shall not include energy capacity augmentations in their proposal. The augmentation strategy will be defined and communicated by the Buyer in subsequent stages. Bidders shall provide detailed technical information on the way the energy capacity augmentations would be implemented during the lifetime of the project.

7. PROPOSAL CONTENT

Bidders shall provide the following documentation as part of their proposal:

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Table 2: Contents of Bidder's Proposal

#	Item	Description	Format
01	Commercial proposal	Single document summarizing the commercial proposal of the Bidder. The LTSA proposal shall align with the principles in Appendix 3.	PDF
02	Bid Forms	Appendix 2 with all related sheets duly populated: BF #1 Technical Parameters BF #2 Performance Parameters BF #3 Scope of Supply BF #4 Certifications BF #5 Track Record BF #6 LTSA BF #7 CAPEX BF #8 Declared Deviations Not complying with this requirement implies rejection of the proposal.	Excel
03	Technical Proposal	Single document explaining the technical solution of the Bidder.	PDF
04	Datasheets	For all equipment listed in Appendix 2 BF #3 (Scope of Supply).	PDF
05	Certifications	Third party certificates listed in Appendix 2 BF #4 (Certifications).	PDF
06	Efficiency Curves	• PCS P-Q curve at different ambient temperatures and output voltages • PCS power – ambient temperature derating curves • LV/MV transformer power – ambient temperature derating curves	PDF
07	Installation Manuals	For all equipment listed in Appendix 2 BF #3 (Scope of Supply).	PDF
08	O&M Manuals	For all equipment listed in Appendix 2 BF #3 (Scope of Supply).	PDF
09	Drawings	• Battery Container drawings • MV Station drawings • BESS Block SLD • Typical foundation drawings	PDF
10	Augmentation	To be provided at a later stage.	PDF
11	Proof of BNEF Tier 1 Status	A valid listing in the latest BNEF Energy Tier 1 report or a certified statement from BNEF.	PDF
12	Report	• Hazard Mitigation Analysis • Large-scale Burn Test Report • Emergency Response Plan • MV, LV and DC cabling sizing report	PDF
13	Disposal Plan	• To ensure all BESS battery modules, upon reaching their end of life (i.e., when they become defective/ non-operational/ non-repairable), are collected, stored, transported, and disposed of or recycled in full compliance with all applicable UAE and international standards, a detailed safe disposal plan, including procedures for interim storage, transport, spill/leakage response, and annual reporting, shall be provided.	PDF

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8. DELIVERY SCHEDULE AND TERMS

Delivery schedule:

- **Delivery:**
 - Phase A COD: by Oct 2027; minimum usable capacity at HV: 1560MWh
 - Phase B COD: by Oct 2028; minimum usable capacity at HV: 3420MWh
 - Phase C (also Plant COD): by Nov 2029; minimum usable capacity at HV: 3420MWh
- **SCOD:**
 - Phase A: by October 2027
 - Phase B: by October 2028
 - Plant (Phase C): by November 2029

The following execution timeline shall also be considered:

- FAT to SAT: 4 months
- SAT to PCOD: 2 months

Bidders shall declare the maximum weekly and monthly production and delivery capacity in Appendix 2 BF #7 (CAPEX).

Bidders shall consider the following delivery terms, according to Incoterms 2020, for the supply from overseas: EXW, FCA, FOB and CIF. Bidders shall clearly identify the address of the manufacturing facilities for different BESS equipment and shall provide a preliminary packing list from each manufacturing facility considered.

9. RFP PLANNING

RFP planning:

Milestone	Estimated Date
Issuance of RFQ Round 1	10/11.2025
Offer submission deadline – Round 1	19/11/2025
Clarifications to bidders- Round 1	20/11/2025- 28/11/2025
Issuance of RFQ Round 2	01/12/2025
Offer submission deadline – Round 2	15/12/2025
Clarifications to bidders- Round 2	16/12/2025- 29/12/2025
Issuance of RFQ Round 3	05/01/2026
Offer submission deadline – Round 3	15/01/2026
Clarifications to bidders- Round 3	16/01/2026-30/01/2026
Shortlisting of the BESS OEM's	05/02/2026
Bid Submission Deadline	20/02/2026
PPA Execution	June 2026
Financial Closure	September 2026
Phase A SCOD	October 2027
Phase B SCOD	October 2028
Plant COD (Phase C SCOD)	November 2029

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10. CONTACT

All proposals, documents, information, comments and requests of any kind concerning this RFP must be communicated to:

Jad Masri
Director, Contracts & Supply Chain
Email: jalmasri@masdar.ae

Marina Molla Domenech
Head, Logistics & Supply Chain
Email: mmolla@masdar.ae

Dinesh Xavier
Senior Manager, Procurement & Logistics
Email: dxavier@masdar.ae

11. CLARIFICATIONS

Conference calls shall be held with Bidders to clarify technical and commercial points. Bidders shall send all clarifications in written prior to the clarification calls.

12. OFFER VALIDITY

The commercial proposal shall remain valid for a period of at least 6 months starting from the issuance date of the Bidders' proposal.

13. PRICES

Prices shall be in US Dollars (USD).

Pricing breakdown shall be provided in Appendix 2 BF #6 (LTSA) and Appendix 2 BF #7 (CAPEX).

14. CONFIDENTIALITY

Bidders shall:

- Keep the information contained within this RFP or learned as part of the RFP process (the "Confidential Information") strictly confidential.
- Restrict the dissemination of Confidential Information on a "need-to-know" basis within its organization.
- Not disclose such Confidential Information to anyone outside its organization without the prior written consent of the Sponsors.
- Not exploit any material information submitted and not use any information or make copies or extracts or summaries thereof for any purpose, except the preparation and submission of the proposal and supporting documents in connection with the RFP and any correspondence, discussions or negotiations with the Buer during the RFP process.

15. APPENDICES

The appendices of this RFP are:

- Appendix 1 – Technical Specification – BESS Main Equipment Requirements

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- Attachment 1: Offtaker MFS
- Attachment 2: IDT and RMU Technical Specification
- Appendix 2 – Bid Forms
- Appendix 3 – LTSA Principles
- Appendix 4 – BESS Compliance Certificate & PCS Compliance Certificate